

CN7023 - Artificial Intelligence & Machine Vision

Week-03 Tutorial Submission (5 Marks)

Task 2: Coursework Project Topic and Dataset Approval

1. Topic of the Coursework Project:

Project Title: Deep Learning-Based Automated Classification of Rice Leaf Diseases

Problem Description & Proposed Solution:

Rice is one of the most critical staple crops globally, supporting billions of people. However, diseases such as Bacterial Leaf Blight, Leaf Blast, Brown Spot, and Leaf Smut pose significant threats to rice yield and agricultural economies. Traditional manual inspection by domain experts is slow, labor-intensive, and prone to error, which can delay disease management. This project proposes a Machine Vision-based automated classification system using deep convolutional neural networks. By capturing digital images of infected rice leaves, the system can automatically classify the leaf into one of nine distinct health/disease classes, enabling rapid, objective, and scalable diagnostics.

Visual Data & Related Challenges:

1. High Intra-class Variation and Inter-class Similarity: Different diseases can exhibit highly similar visual symptoms (e.g., small brown spots vs. early leaf blast spots), while the same disease can manifest differently depending on the stage of infection.
2. Environmental and Background Noise: Field-captured images contain varying lighting conditions, shadow effects, and background clutter (like soil, water, and weeds), which can interfere with feature extraction.
3. Class Imbalance: The dataset contains a high number of samples for some classes (2,000 images) and very few for others (approx. 430 images), requiring robust regularization and class balancing techniques.

2. Dataset Details Table:

Dataset Name	RICE Leaf Diseases
Dataset Link / URL	https://www.kaggle.com/datasets/yusufmurtaza01/rice-leaf-diseases
Dataset Size	486 MB
Image Dimensions	587 x 587 pixels
Number of Classes	9 Classes
Number of Images per Class	Class 0 (Rice__BacterialLeafBlight): 2,000 images Class 1 (Rice__BrownSpot): 2,000 images Class 2 (Rice__Healthy): 433 images Class 3 (Rice__Hispa): 431 images Class 4 (Rice__LeafBlast): 449 images Class 5 (Rice__LeafScald): 456 images Class 6 (Rice__LeafSmut): 2,000 images Class 7 (Rice__NarrowBrownLeafSpot): 443 images Class 8 (Rice__NeckBlast): 453 images Total Images: 8,665
Notes (Additional Info)	The dataset is highly imbalanced between classes. To train a robust classifier, techniques such as focal loss, class-weighted cross-entropy, and extensive data augmentation (rotation, shearing, color jitter) will be implemented.

Task 3 & 4: Literature Review and Proposed Methods

Reviewing the relevant literature using Google Scholar, we have identified the top four appropriate deep learning methods suitable for solving the rice leaf disease classification problem:

Method 1: Convolutional Neural Networks (CNNs) with Custom Feature Extraction

Description: Custom CNN architectures are designed to extract hierarchical spatial features (like spots, patterns, and lesions) directly from raw images using convolutional filters. They are highly effective for capturing local patterns characteristic of specific plant diseases.

Paper Source: Analysis of Convolutional Neural Network-based Image Classifications: A Multi-Featured Application for Rice Leaf Disease Prediction and Recommendations for Farmers

Link: <https://arxiv.org/abs/2410.01827>

Method 2: Transfer Learning with Pre-trained Deep CNNs (ResNet / MobileNet)

Description: Using pre-trained weights from large-scale image databases (e.g., ImageNet) and fine-tuning the model on the rice leaf disease dataset. This approach accelerates convergence and helps mitigate overfitting, particularly for classes with fewer samples.

Paper Source: Leveraging Pre-trained CNNs for Efficient Feature Extraction in Rice Leaf Disease Classification

Link: <https://arxiv.org/abs/2405.00025>

Method 3: Real-Time Object Detection and Localized Classification (YOLOv8)

Description: Utilizing the state-of-the-art YOLOv8 architecture to perform local boundary box detection and classification of disease regions simultaneously, optimizing processing speed for potential mobile/edge device deployments in agricultural fields.

Paper Source: Deep Learning for the Classification of Rice Leaf Diseases Using YOLOv8

Link: <https://doi.org/10.18280/mme.370414>

Method 4: Explainable AI (XAI) and Model Interpretability

Description: Integrating models with visual explanation techniques like Grad-CAM and LIME. This highlights the specific pixels or features (such as leaf lesions) that influenced the model's prediction, ensuring diagnostic trustworthiness for agronomic application.

Paper Source: Exploring Deep Learning and Explainable AI for Precise Rice Leaf Disease Detection and Classification

Link: <https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A1838634>